25901445 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY207 - CURRENT ELECTRICITY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. State the sign convention used while applying Kirchhoff's Voltage Law. [U] / [CO1]
2. What are the factors that affect the resistance of a conductor? [U] / [CO1]
3. What is the condition for maximum power transfer in a circuit? [U] / [CO2]
4. Define electric current? What is its unit? [K] / [CO1]
5. What are the practical units of capacitance? [U] / [CO3]
6. Explain Faraday’s law of electromagnetic induction. [U] / [CO3]
7. Describe a variable capacitor. [K] / [CO3]
8. How does the separation between the plates affect the the capacitance of a parallel plate capacitor? [U] / [CO3]
9. Briefly explain electrolytic dissociation. [K] / [CO4]
10. Define phase angle in RC circuit. [U] / [CO5]

11. Write expression for impedance in series LCR circuit. [U] / [CO5]

12. Define resonance in an LCR circuit. [U] / [CO5]

13. Why does an emf develop when two dissimilar metals are joined and junctions are kept at different temperatures? [An] / [CO6]

14. Define Thomson effect. [K] / [CO6]

[2x10 = 20]

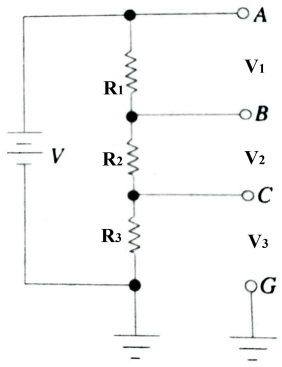
Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. Find the expression for V_1 , V_2 and V_3 of the voltage divider circuit shown in the figure. [U] / [CO1]



16. State and explain the Superposition theorem. [U] / [CO2]

17. An iron ring of 40 cm mean radius made of a square iron bar of cross section area 5cm^2. A coil of 500 turns is uniformly wound over the iron ring with a wire of 0.003 cm^2 cross section area. If relative permiability of iron is 6000, calculate the self inductance of the coil. [A] / [CO3]

18. Derive the expression for coefficient of mutual inductance in terms of the dimensions of the coils. [U] / [CO3]

19. Show that the power consumed per cycle is zero for a circuit containing inductance only. [A] / [CO5]

20. What is skin effect? How does it affect transformer losses? [An] / [CO5]

21. What is the time constant in LR circuit? Illustrate the variation of current with time during the decay process. [An] / [CO5]

22. Obtain equation for capacitor voltage during charging and plot graph. [A] / [CO5]

[5x6 = 30]